1. Sample 1 of water froze at 0°C.
2. Sample 2 of water froze at 0°C.
3. Sample 3 of water froze at 0°C.

.....

N. Sample N of water froze at 0°C.

N+1. If all past samples of water froze at 0°C, then this sample of water will freeze at 0°C.

———————————————————-

C. This sample of water will freeze at 0°C. (1-N+1)

# Is this argument valid?

We already have an explanation of why we should believe premises 1-N.
What about premise N+1?

In our first reading, Hume addresses the question of whether we should believe premises like this by drawing a distinction between two different kinds of claims:

Premise N+1 appears to be like the claim that the sun will rise tomorrow: it is a matter of fact rather than a matter of the relations of ideas, and so cannot be known “by the mere operation of thought.”

N+1. If all past samples of water froze at 0°C, then this sample of water will freeze at 0°C.

## The Uniformity of Nature

The future will be like the past.

It seems as though, if we should believe in the Uniformity of Nature, we should believe N+1. So the basic question is whether we should believe in the Uniformity of Nature.

Is the negation of this claim a contradiction?

It seems not. So it seems that, if we should believe it, we must believe it on the basis of experience.

After all, yesterday the future was like the past. And the same for the day before that. And this suggests an argument for the Uniformity of Nature.

1. Yesterday, the future was like the past.
2. The day before yesterday, the future was like the past.
3. The day before the day before yesterday, the future was like the past.

.....

N. N days ago, the future was like the past.

———————————————-

C. Today, the future will be like the past. (1-N)

It is hard to see how we could make the argument valid without adding a premise which was just a restatement of the very claim — the Uniformity of Nature — which we were trying to prove.

All massive objects attract one another.

Every 24 hours, the earth
rotates on its axis.

It is very plausible that these are all claims which we should believe. But we have at present no explanation of this fact. After all, the only arguments we can give for these claims will be arguments like our argument that the sun will come up tomorrow. And these arguments are invalid, and hence not good arguments. This seems to show that Foundationalism, as we have formulated it, is false.

Halley’s comet will next be visible from earth in 2061.

It also raises an important question. We think that at least some scientific theories are theories we should believe, and that other generalizations are not. Consider, for example, People born under the sign of Taurus are more likely to be stubborn than people born under other signs.

This is is a prediction of the theory of astrology. One shouldn't believe this prediction because one shouldn't believe the theory. But why not?

It is tempting to say something like: there's no good argument for astrology; there's no reason to think that it is true.

That leads to our central question: what does it mean for a theory to be **well-supported** by the evidence?

The examples we have already discussed give us a plausible answer to this question. Scientific theories typically involve certain generalizations. In the simplest case, they will be claims of the form

All F’s are G.

These are not claims which our senses can tell us to be true. But our senses can tell us that claims like this are true:

This particular thing is an F, and it is G.

Let's call claims which are related in this way to generalizations **instances** of the generalization.

Then we might say that a generalization is well-supported by the evidence just in case the following two conditions are satisfied:

Our senses tell us that many instances of the generalization are true.

Our senses don’t tell us that any instances of the generalization are false.

## Induction → Belief

If you know many true instances of a generalization P, and don't know of any false instances of P, you should believe P.

I want to now look at two problems for Induction $\rightarrow$ Belief.

The first is called **the paradox of the ravens**. Consider the following generalization:

All ravens are black.

All non-black things are non-ravens.

Here's a second investigation I could undertake:

Suppose that I now begin to investigate my immediate environment for non-black things, checking whether they are ravens. I check my shirt, the lectern, my laptop — and it turns out that none of them are ravens.

## Investigation #2

I begin to investigate my immediate environment. I check the first ten non-black things I can find — and it turns out that none of them are ravens.

Here, as in the previous generalization, I have found ten true instances of one of the two generalizations. So it looks as though if Induction → Belief is true, I should substantially increase my confidence in the claim that all ravens are black.

But intuitively, this is bizarre. Surely the fact that a bunch of non-black things in my environment are also non-ravens should do nothing, or almost nothing, to support the generalization that all ravens are black.

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